



Research article

Mechanism of METTL3 in the proliferation, invasion, and migration of intrahepatic cholangiocarcinoma cells via m6A modification

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ABSTRACT

Intrahepatic cholangiocarcinoma (ICC) is a primary invasive malignant tumor. This study was conducted to explore the role of methyltransferase-like 3 (METTL3)-mediated m6A modification in ICC cells and provide novel targets for ICC treatment. Levels of METTL3/YTH N6-methyladenosine RNA binding protein 2 (YTHDF2)/Nedd4 family interacting protein 1 (NDFIP1) in cells were determined. Cell viability, proliferation, invasion, and migration were evaluated. The enrichments of METTL3, YTHDF2, and m6A on NDFIP1 mRNA were analyzed. The mRNA stability was determined. Inhibition of YTHDF2 or NDFIP1 was combined with si-METTL3 to confirm the mechanism. The role of METTL3 *in vivo* was verified. METTL3 was overexpressed in ICC cells. METTL3 silencing suppressed ICC cell malignant behaviors, which were reversed by METTL3 overexpression. METTL3 increased m6A modification on NDFIP1 mRNA, facilitated YTHDF2 recognition of m6A, and promoted NDFIP1 mRNA degradation, thereby suppressing NDFIP1 expression. YTHDF2 inhibition increased NDFIP1 mRNA levels. NDFIP1 downregulation partially reversed the inhibitory effects of si-METTL3 on ICC cell behaviors, while NDFIP1 overexpression partially reversed the promotive effects of METTL3 on ICC cell behaviors. METTL3 downregulation suppressed ICC growth by increasing NDFIP1 expression. In conclusion, METTL3 aggravates ICC cell proliferation, invasion, and migration by promoting the degradation of NDFIP1 mRNA in a YTHDF2-dependent manner.

1. Introduction

Intrahepatic cholangiocarcinoma (ICC), a primary malignancy of the liver, arises invasively from the epithelial lining of the intrahepatic biliary tree [1]. ICC accounts for approximately 10–15 % of primary liver cancer cases worldwide, and there were over 90,000 new cases and 83,000 deaths attributed to ICC in 2020 [2]. Risk factors for ICC include liver cirrhosis, viral hepatitis, and unhealthy lifestyle habits [3]. Approximately 70 % of ICC patients are not eligible for surgical resection and require systemic treatment, and the efficacy of the first-line regimen, the combination of chemotherapy, gemcitabine, and cisplatin, remains limited [4]. Moreover, even among patients who undergo curative surgical treatment, the five-year overall survival rate is only 30 % [5]. Due to the limited knowledge of effective treatment methods for ICC and its poor prognosis, there is an urgent need to understand the mechanisms driving the progression of ICC.

Dysregulated epigenetic mechanisms leading to aberrant genetic

expression are intimately linked with the onset and progression of ICC [6]. N6-methyladenosine (m6A) is a specific epigenetic modification and the most prevalent and abundant modification in eukaryotic RNA [7]. m6A modification is a dynamic and reversible process regulated by "writers" and "erasers" and recognized by "readers" [8]. Methyltransferase-like 3 (METTL3), as a core component of the m6A methyltransferase complex, dynamically regulates the expression of oncogenes/tumor suppressor genes through m6A modification to participate in cancer progression [9–11]. Importantly, increased evidence suggests that METTL3 plays an oncogenic role in ICC via m6A modification [12,13]. YTH N6-methyladenosine RNA binding protein 2 (YTHDF2), functioning as an m6A reader, specifically identifies and interacts with m6A-modified sites within cytoplasmic mRNA, thereby promoting the expedited decay of its targeted transcripts [14]. Reports indicate that METTL3 induces high levels of mRNA m6A modification in cancer cells and reduces the mRNA stability and expression of target genes in a YTHDF2-dependent manner [15,16]. Based on these reports,

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our study further investigated the mechanism of METTL3 and YTHDF2 in ICC.

Nedd4 family interacting protein 1 (NDFIP1) is an E3 ubiquitin ligase of the NEDD4 family and is involved in tumor suppression [17]. Extensive research findings have confirmed that NDFIP1 is a tumor suppressor. For instance, NDFIP1 can inhibit tumor growth and cancer cell proliferation in non-small cell lung cancer [18]. Overexpression of NDFIP1 inhibits cell proliferation, migration, and invasion in hepatocellular carcinoma [19]. Of note, reducing the protein levels of NDFIP1 through ubiquitination can promote the progression of ICC [20]. Database predictions have revealed the presence of m6A modifications on NDFIP1 mRNA. We speculated that NDFIP1 may act as a downstream mechanism for METTL3 in ICC via m6A modification.

In our study, we analyzed the aberrant expression of METTL3 in ICC and investigated the mechanism of METTL3 in regulating NDFIP1 expression in an m6A-YTHDF2-dependent manner, providing new theoretical knowledge for the treatment of ICC.

2. Materials and methods

2.1. Ethics statement

The animal experiments were conducted in accordance with the guidelines provided by the Animal Care and Use Committee of The First Affiliated Hospital of University of South China. All procedures involving the use, care, and handling of animals followed the Guide for the Care and Use of Laboratory Animals by the National Institutes of Health of the United States [21].

2.2. Cell culture

The human normal intrahepatic bile duct epithelial cell line (HIBEpic) and ICC cell lines (HUCCT1, RBE, HCCC-9810) were provided by the Type Culture Collection of the Chinese Academy of Science. All cells were cultured in Roswell Park Memorial Institute (RPMI)-1640 medium (Sigma-Aldrich, St. Louis, MO, USA) supplemented with 10 % fetal bovine serum (FBS, Invitrogen, Waltham, MA, USA), 1 % streptomycin, and 1 % penicillin (Solarbio, Beijing, China). The cells were maintained in a humidified atmosphere at 37 °C with 5 % CO₂.

2.3. Cell treatment

The small interfering RNAs (siRNAs) targeting human METTL3, YTHDF2, and NDFIP1, as well as the non-targeting control siRNAs, were purchased from RiboBio Co., Ltd. (Guangzhou, Guangdong, China). Full-length METTL3 or NDFIP1 was subcloned into the pcDNA3.1 vector (GenePharma, Shanghai, China), and empty pcDNA3.1 vector (GenePharma) was used as a control. When the confluence of HUCCT1 or HCCC-9810 cells reached 80 %, the cells were transfected using Lipofectamine 2000 (Invitrogen, Carlsbad, CA, USA). To obtain cells with stable METTL3 expression, sh-METTL3/sh-NDFIP1 or oe-METTL3/oe-NDFIP1 was packaged into lentiviruses (GenePharma) and then used to infect HUCCT1 or HCCC-9810 cells. Cells with stable METTL3 expression were selected by treatment with puromycin (2 µg/mL) and used for *in vivo* experiments.

2.4. Cell counting kit-8 (CCK-8) method

CCK-8 kit (Sigma-Aldrich) was used to assess cell proliferation. HUCCT1 or HCCC-9810 cells (3×10^3 cells/well) were seeded in 96-well plates. After incubation, 10 µL of CCK-8 reagent was added to each well for 2-h incubation in the dark. Subsequently, the absorbance at a wavelength of 450 nm was monitored daily at 24-h intervals employing a microplate reader (BioTek, Winooski, Vermont, USA).

2.5. 5-ethynyl-2'-deoxyuridine (EdU) staining

For the EdU staining assay, HUCCT1 or HCCC-9810 cells were cultured in 96-well plates for 24 h. After incubation with a 50 mM EdU solution for 2 h, the cell lines were fixed with 4 % paraformaldehyde. Subsequently, the cells were sealed with Apollo dye solution and Hoechst 33342. The EdU-labeled cells were observed and measured using a microscope (Olympus, Tokyo, Japan).

2.6. Transwell assay

Transwell assays were performed to assess the migratory and invasive abilities of HUCCT1 or HCCC-9810 cells. Transwell assay was conducted using 24-well plates equipped with chamber inserts containing 8.0 µm pores (Corning Incorporated, Corning, NY, USA). The chamber inserts were either left empty or filled with 2 % Matrigel (BD Science, San Jose, CA, USA). The cells were trypsinized and resuspended in serum-free culture medium. Subsequently, 2×10^4 cells were introduced into each well of the upper compartments. The lower chamber was filled with a cell-nutrient inducer composed of 600 µL of medium supplemented with 20 % FBS. After incubation at 37 °C for 24 h, cells were carefully extracted using a cotton swab. To make the migrated or invaded cells visible, they were fixed with 4 % paraformaldehyde and stained with 0.5 % crystal violet for 30 min. Subsequently, the migrated and invaded cells were observed and measured using an inverted microscope (Olympus), with 5 random microscopic fields selected for analysis.

2.7. m6A quantification

The m6A RNA methylation quantification kit (Colorimetric) (ab185912, Abcam, Cambridge, MA, USA) was used to detect the total m6A in cells according to the manufacturer's instructions. In brief, total RNA was extracted and 200 ng of RNA was incubated with 80 µL of binding solution at 37 °C for 90 min. Subsequently, the diluted capture antibody and detection antibody were added for incubation. After thorough washing, the developing solution was added and the absorbance was measured at 450 nm using a microplate reader within 15 min.

2.8. Actinomycin D treatment

HUCCT1 or HCCC-9810 cells were seeded in a 6-well plate and treated with actinomycin D (5 µg/mL, Sigma-Aldrich) for a specified period. Total RNA was extracted using TRIzol reagent and analyzed by reverse transcription quantitative polymerase chain reaction (RT-qPCR).

2.9. RNA immunoprecipitation (RIP) and methylated RNA immunoprecipitation (MeRIP) assays

RIP assay was performed using the Magna RIP™ kit (Millipore, Billerica, MA, USA) according to the protocol. Magnetic beads were coated with antibodies against immunoglobulin G (IgG) (ab172730, Abcam), METTL3 (ab195352, Abcam), and YTHDF2 (ab220163, Abcam) at room temperature for 30 min. To capture NDFIP1, cell lysates from HUCCT1 or HCCC-9810 cells (2×10^7) were incubated with the antibody-coated magnetic beads.

For MeRIP, TRIzol was used to extract the total RNA. Similar to RIP assay, magnetic beads (Millipore) were coated with 5 µg of an anti-IgG or anti-m6A antibody (ab208577, Abcam) and incubated at room temperature for 30 min. Subsequently, 50 µg of total RNA was added to the antibody-coated beads and incubated overnight at 4 °C in immunoprecipitation buffer containing an RNase inhibitor. After proteinase K digestion, the m6A-bound RNA was precipitated using the RNA Clean & Concentrator kit (ZYMO research, Hangzhou, China). The level of m6A enrichment in NDFIP1 was assessed using qPCR and normalized to the input. For MeRIP of tissue samples, the tumor tissues were first ground

Table 1
PCR primer sequences.

Gene	Sequences (5'-3')
METTL3	F: TGGGGGTATGAACGGGTAGA R: TGGTTGAAGCCTTGGGGATT
YTHDF2	F: CAGGCATCAGTAGGGCAACA R: TTATGACCGAACCCTGCTCC
NDFIP1	F: TACTGGCTCTGGTGGGTGTT R: ACTCTGGTCTCTGGGAGATT
GAPDH	F: GATGCTGGCGCTGAGTACG R: GCTAAGCAGTTGGTGGTGC

with liquid nitrogen, followed by addition of lysis buffer (containing protease inhibitors and RNase inhibitors) for lysis, and then the above-mentioned operations were carried out.

2.10. Dual-luciferase assay

The m6A modification site of NDFIP1 mRNA was predicted from the SRAMP database (<http://www.cuilab.cn/sramp/>) [22]. The 3'UTR region of NDFIP1 was amplified from complementary DNA (cDNA) and cloned into the psicheck-2 vector to construct the WT-NDFIP1 dual

luciferase reporter plasmid (GGACU). The m6A site on the 3'UTR of NDFIP1 was mutated (GGCCU) by Sangon Biotech (Shanghai, China) and then cloned into the psicheck-2 vector to construct the MUT-NDFIP1 dual luciferase reporter plasmid. For the dual-luciferase reporter gene experiment, HUCCT1 or HCCC-9810 cells were cultured in the 6-well plate and transfected with the reporter plasmid and YTHDF2 pcDNA3.1 or empty pcDNA3.1. At 48-h, luciferase activity was measured using the dual luciferase reporter assay system (Promega, Madison, WI, USA) according to the manufacturer's instructions.

2.11. Xenograft tumor experiment in nude mice

BALB/c nude mice (male, 4 weeks old, weighing 15–20 g) were obtained from Beijing Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). Mice were housed in a specific pathogen-free animal facility with a 12-h light-dark cycle. Lentivirus-infected HUCCT1 or HCCC-9810 cells (4×10^6) were suspended in 0.1 mL of phosphate-buffered saline and were subcutaneously injected into the right flank of nude mice. Tumor size was monitored every 4 days for a total of 28 days, and tumor volume was calculated as length $\times$ width²/2. After 4 weeks, the mice were euthanized by intraperitoneal injection of 200 mg/kg sodium pentobarbital. Then, tumor tissues were excised and weighed

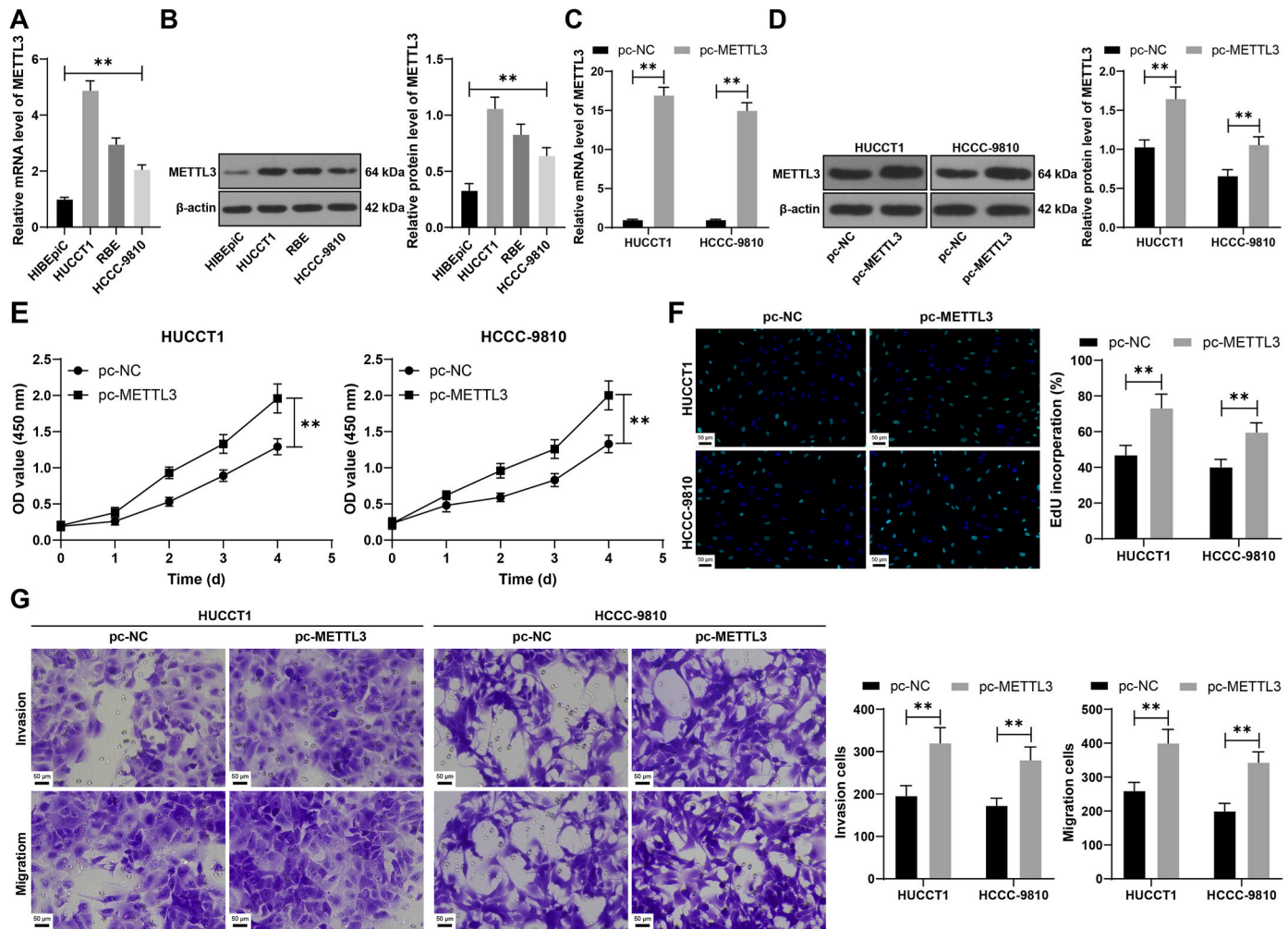


Fig. 1. METTL3 promotes proliferation, invasion, and migration of ICC cells. A–B: The expression of METTL3 in different cells was detected by RT-qPCR and Western blot assays. pc-METTL3 was transfected into HUCCT1 and HCCC-9810 cells, with pc-NC as the negative control. C–D: The expression of METTL3 in cells was detected by RT-qPCR and Western blot assays. E: Cell proliferation was measured using the CCK-8 assay. F: The number of EdU-positive cells was detected by EdU staining. G: Cell invasion and migration were assessed using the Transwell assay. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. One-way ANOVA was used for data comparison among multiple groups in panels A–B, while two-way ANOVA was used for data comparison among multiple groups in panels C–G. Tukey's multiple comparisons test was used for post hoc analysis. ** $p < 0.01$.

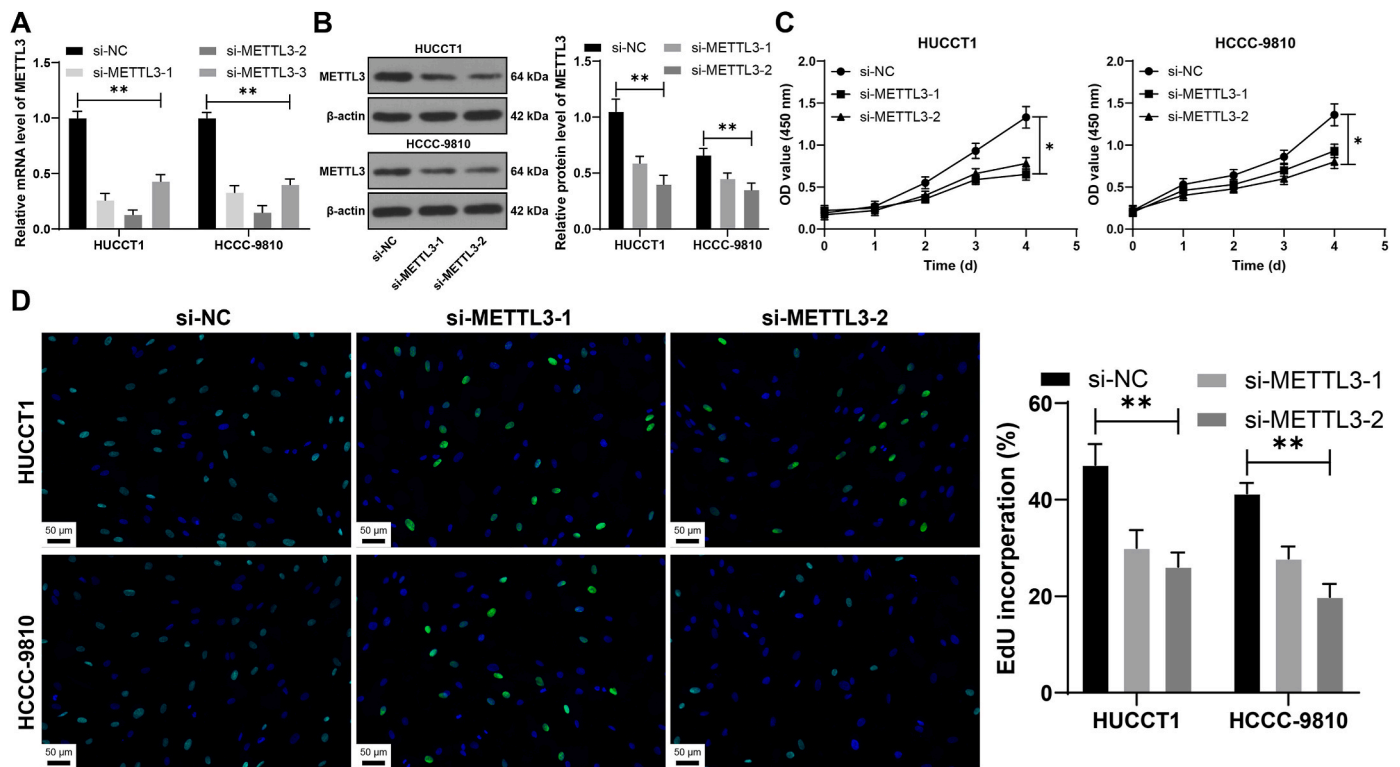


Fig. 2. Downregulation of METTL3 inhibits proliferation, invasion, and migration of ICC cells. HUCCT1 and HCCC-9810 cells were transfected with si-METTL3, with si-NC transfected as the negative control. A–B: The expression of METTL3 in cells was detected by RT-qPCR and Western blot assays. C: Cell proliferation was measured using the CCK-8 assay. D: The number of EdU-positive cells was detected by EdU staining. E: Cell invasion and migration were assessed using the Transwell assay. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. Two-way ANOVA was used for data comparison among multiple groups, and Tukey's multiple comparisons test was used for post hoc analysis. * $p < 0.05$, ** $p < 0.01$.

for further analysis.

2.12. Immunohistochemistry assay

Tumor tissues were fixed in 4 % formaldehyde solution, dehydrated in gradient ethanol, and embedded in paraffin. After antigen retrieval, the tissue sections were incubated in a H_2O_2 solution for 10 min. After washing, the sections were incubated with the Ki-67 antibody (ab16667, Abcam) for 16 h, followed by incubation with a secondary antibody (ab6721, Abcam). The signal was visualized by adding a DAB substrate. After washing, the sections were stained with hematoxylin and imaged using a BX51 microscope.

2.13. RT-qPCR

Total RNA was extracted from cells using TRIzol reagent (Invitrogen) according to the manufacturer's instructions. The integrity and purity of the extracted RNA were evaluated using a NanoDrop One ultramicro UV spectrophotometer. Subsequently, cDNA was synthesized from the total RNA using the PrimeScript RT Reagent Kit (Accurate Biology, Changsha, China) on a qPCR instrument. Real-time quantitative PCR was carried out using SYBR® Premix Ex Taq™ II (Takara, Shiga, Japan) on an ABI 7900HT system (Applied Biosystems, Foster City, CA, USA) to assess the expression levels of target genes. The PCR primers used are shown in Table 1. The expression levels were calculated using the $2^{-\Delta\Delta Ct}$ method [23], with glyceraldehyde-3-phosphate dehydrogenase (GAPDH) as the reference gene.

2.14. Western blot analysis

Total protein was extracted from cells using

radioimmunoprecipitation assay buffer containing protease and phosphatase inhibitors. The protein concentration was determined using the bicinchoninic acid assay kit (Beyotime, Shanghai, China). The proteins were separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis and then transferred onto a polyvinylidene fluoride membrane (Millipore). The membrane was blocked with 5 % skim milk for 1.5 h and then incubated overnight at 4 °C with primary antibodies against METTL3 (ab195352, 1:1000, Abcam), YTHDF2 (ab220163, 1:1000, Abcam), NDFIP1 (ab236892, 1:1000, Abcam), and β -actin (ab8227, 1:1000, Abcam). Afterwards, the membrane was incubated with a secondary antibody (ab205718, 1:2000, Abcam) at 37 °C for 2 h. The target proteins were visualized using an enhanced chemiluminescence reagent (Millipore). β -actin was used as a loading control.

2.15. Statistical analysis

SPSS 21.0 (IBM, Armonk, NY, USA) and GraphPad Prism 8.0 software (GraphPad Software Inc., San Diego, CA, USA) were used for statistical analysis and data plotting. First, normality and homogeneity of variance tests were conducted, which verified that the data were in normal distribution and homogeneity of variance. Data comparison between two groups was performed using *t*-test. Data comparisons among multiple groups were performed using one-way or two-way analysis of variance (ANOVA), followed by Tukey's multiple comparisons test for post hoc analysis. *p* value was obtained via two-sided tests; $p < 0.05$ indicated a statistically significant difference and $p < 0.01$ indicated a highly significant difference.

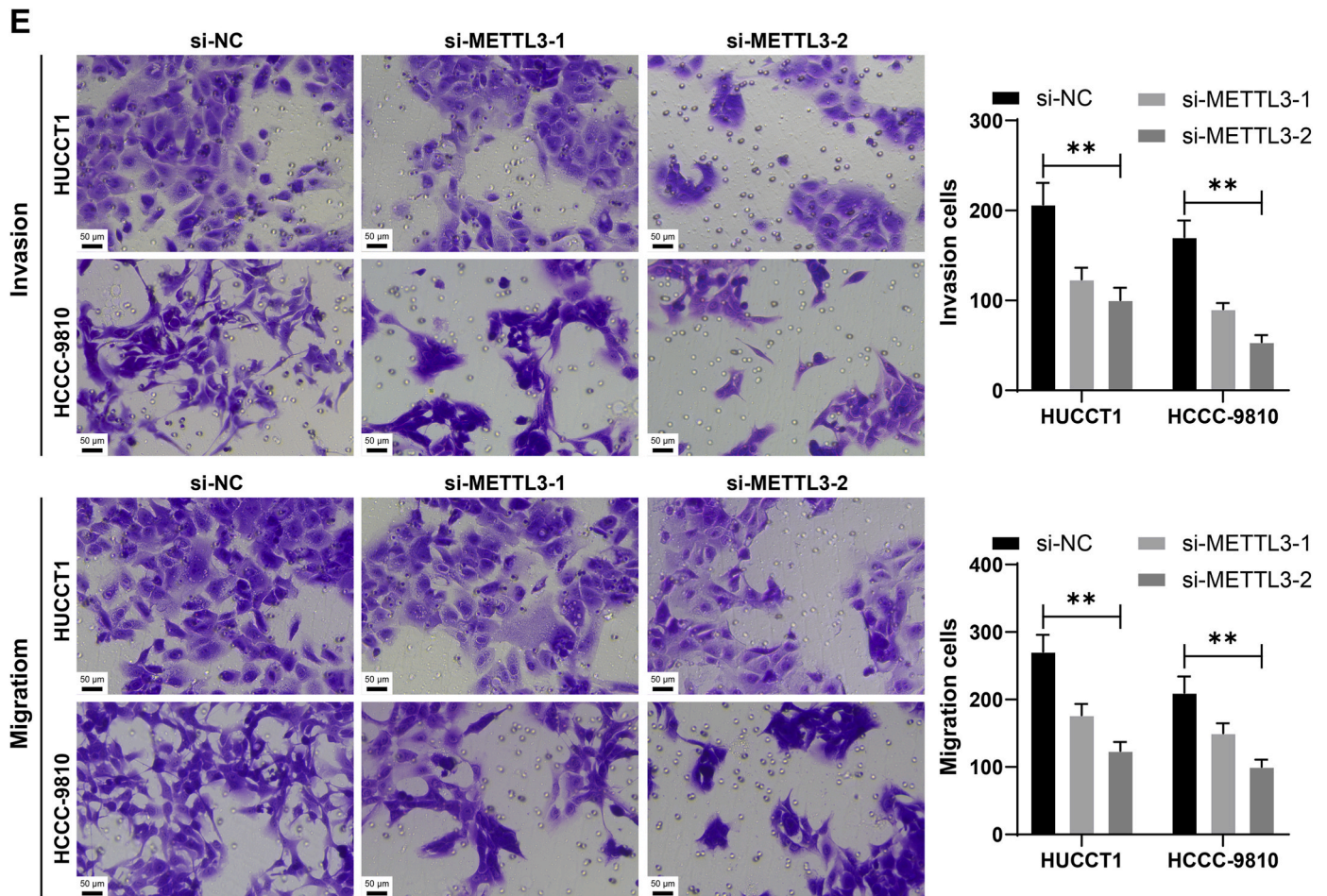


Fig. 2. (continued).

3. Results

3.1. METTL3 promotes proliferation, invasion, and migration of ICC cells

METTL3 is highly expressed in ICC [12,13]. We examined the expression of METTL3 in various cell lines using RT-qPCR and Western blot assays. The results showed that compared to HIBEpC cells, METTL3 expression in ICC cell lines (HUCCT1, RBE, HCCC-9810) significantly increased ($p < 0.05$, Fig. 1A–B). HUCCT1 cells with relatively high expression and HCCC-9810 cells with relatively low expression of METTL3 were selected for further experiments. Transfection of pc-METTL3 into cells significantly upregulated METTL3 expression in both cell lines ($p < 0.05$, Fig. 1C–D), along with remarkable increases in cell proliferation and EdU-positive cells ($p < 0.05$, Fig. 1E–F). Moreover, overexpression of METTL3 promoted the number of invading and migrating ICC cells ($p < 0.05$, Fig. 1G). Taken together, these findings suggest that METTL3 is highly expressed in ICC and promotes proliferation, invasion, and migration of ICC cells.

3.2. Downregulation of METTL3 inhibits proliferation, invasion, and migration of ICC cells

Subsequently, we transfected HUCCT1 and HCCC-9810 cells with si-METTL3 to downregulate METTL3 expression ($p < 0.05$, Fig. 2A–B) and selected the two siRNAs (si-METTL3-1 and si-METTL3-2) with better transfection efficiency for subsequent experiments. Compared to the si-NC group, cells in the si-METTL3 group showed reduced proliferation and decreased EdU-positive cells ($p < 0.05$, Fig. 2C–D). Furthermore, downregulation of METTL3 decreased the number of invading and

migrating ICC cells ($p < 0.05$, Fig. 2E). These results suggest that downregulation of METTL3 can inhibit proliferation, invasion, and migration of ICC cells.

3.3. METTL3 increases m6A modification on NDFIP1 mRNA and decreases NDFIP1 mRNA stability

METTL3 mediates m6A modification on IFIT2 mRNA, accelerating IFIT2 mRNA decay [12]. The RPISeq database (<http://pridb.gdcbio.iastate.edu/RPISeq/>) [22] predicted a high binding probability between METTL3 and NDFIP1 mRNA (Fig. 3A). The SRAMP database (<http://www.cuilab.cn/sramp/>) [22] predicted the presence of m6A modification on NDFIP1 mRNA (Fig. 3B). Additionally, evidence from the literature [20] supports a reduced expression of NDFIP1 in ICC. In alignment with these findings, our data revealed a statistically significant downregulation of NDFIP1 in ICC cell line models ($p < 0.05$, Fig. 3C–D). We hypothesized that NDFIP1 might be a downstream mechanism of METTL3. m6A quantification analysis showed that the m6A content was significantly increased in ICC cell lines compared to HIBEpC cells. Additionally, the m6A content was significantly increased in cells of the pc-METTL3 group compared to the pc-NC group; while the m6A content was decreased in cells of the si-METTL3 group compared to the si-NC group ($p < 0.05$, Fig. 3E–F). Furthermore, overexpression of METTL3 resulted in increased enrichment of METTL3 and m6A modification on NDFIP1 mRNA, whereas downregulation of METTL3 showed the opposite effect ($p < 0.05$, Fig. 3G–H). However, overexpression of METTL3 decreased the mRNA stability and expression of NDFIP1, while downregulation of METTL3 increased the mRNA stability and expression of NDFIP1 ($p < 0.05$, Fig. 3I–K). In conclusion, METTL3 can

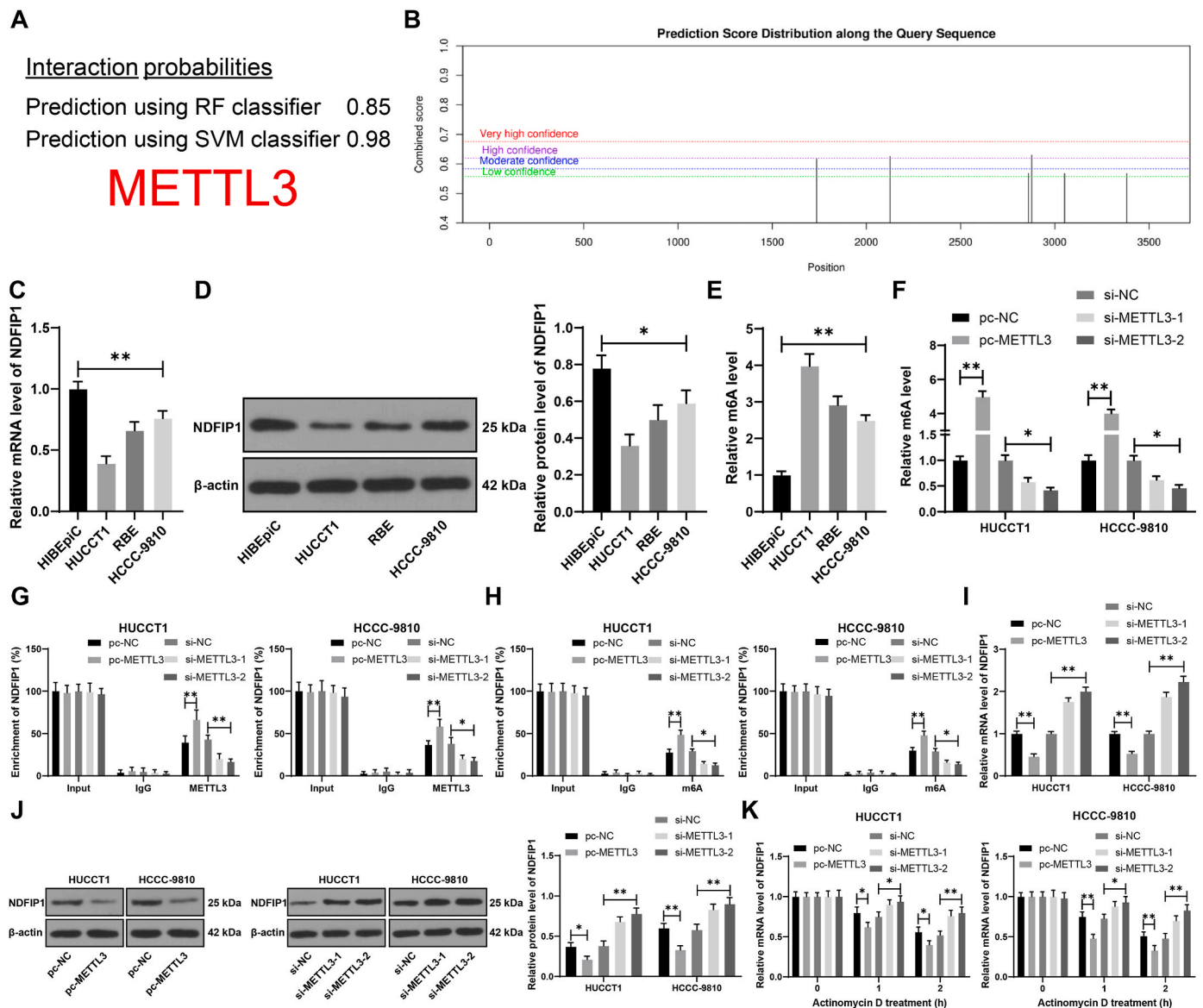


Fig. 3. METTL3 increases m6A modification on NDFIP1 mRNA and decreases NDFIP1 mRNA stability. **A:** The binding probability between METTL3 and NDFIP1 mRNA was predicted by the RPISeq database. **B:** NDFIP1 m6A modification was predicted by the SRAMP database. **C–D:** The expression of NDFIP1 in different cells was detected by RT-qPCR and Western blot assays. **E–F:** Quantitative analysis of m6A levels in cells. **G–H:** RIP analysis of METTL3 enrichment and m6A modification enrichment on NDFIP1 mRNA in cells. **I–J:** The expression of NDFIP1 in cells was detected by RT-qPCR and Western blot assays. **K:** After actinomycin D treatment, the stability of NDFIP1 mRNA in cells was assessed by RT-qPCR. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. One-way ANOVA was used for data comparison among multiple groups in panels C–E, while two-way ANOVA was used for data comparison among multiple groups in panels F–K. Tukey's multiple comparisons test was used for post hoc analysis. * $p < 0.05$, ** $p < 0.01$.

increase m6A modification on NDFIP1 mRNA and decrease NDFIP1 mRNA stability.

3.4. YTHDF2 recognizes m6A and degrades mRNA to suppress NDFIP1 expression

METTL3 activates YTHDF2 to recognize m6A modifications in mRNA [24]. The RPISeq database predicted a high binding probability between YTHDF2 and NDFIP1 mRNA (Fig. 4A). RIP analysis revealed that overexpression of METTL3 increased the enrichment of YTHDF2 on NDFIP1 mRNA, while downregulation of METTL3 decreased the enrichment ($p < 0.05$, Fig. 4B). Subsequently, we downregulated YTHDF2 expression in two cell lines ($p < 0.05$, Fig. 4C–D), which resulted in reduced enrichment of YTHDF2 on NDFIP1 mRNA ($p < 0.05$, Fig. 4B). Downregulation of YTHDF2 promoted NDFIP1 expression ($p <$

0.05, Fig. 4D–E) and increased the stability of NDFIP1 mRNA ($p < 0.05$, Fig. 4F). The dual-luciferase assay confirmed that YTHDF2 can bind to methylated NDFIP1 mRNA ($p < 0.05$, Fig. 4G). In summary, YTHDF2 recognizes m6A modifications and degrades mRNA to suppress NDFIP1 expression.

3.5. Downregulation of NDFIP1 alleviates the inhibitory effects of METTL3 silencing on ICC cell proliferation, invasion, and migration

Subsequently, we successfully downregulated NDFIP1 expression by transfecting si-NDFIP1 and then performed combined experiments using si-NDFIP1-1 and si-METTL3-2, which showed better transfection efficiency ($p < 0.05$, Fig. 5A–B). The results demonstrated that compared to the group with METTL3 silencing alone, the combined group exhibited increased cell proliferation and increased numbers of invading and

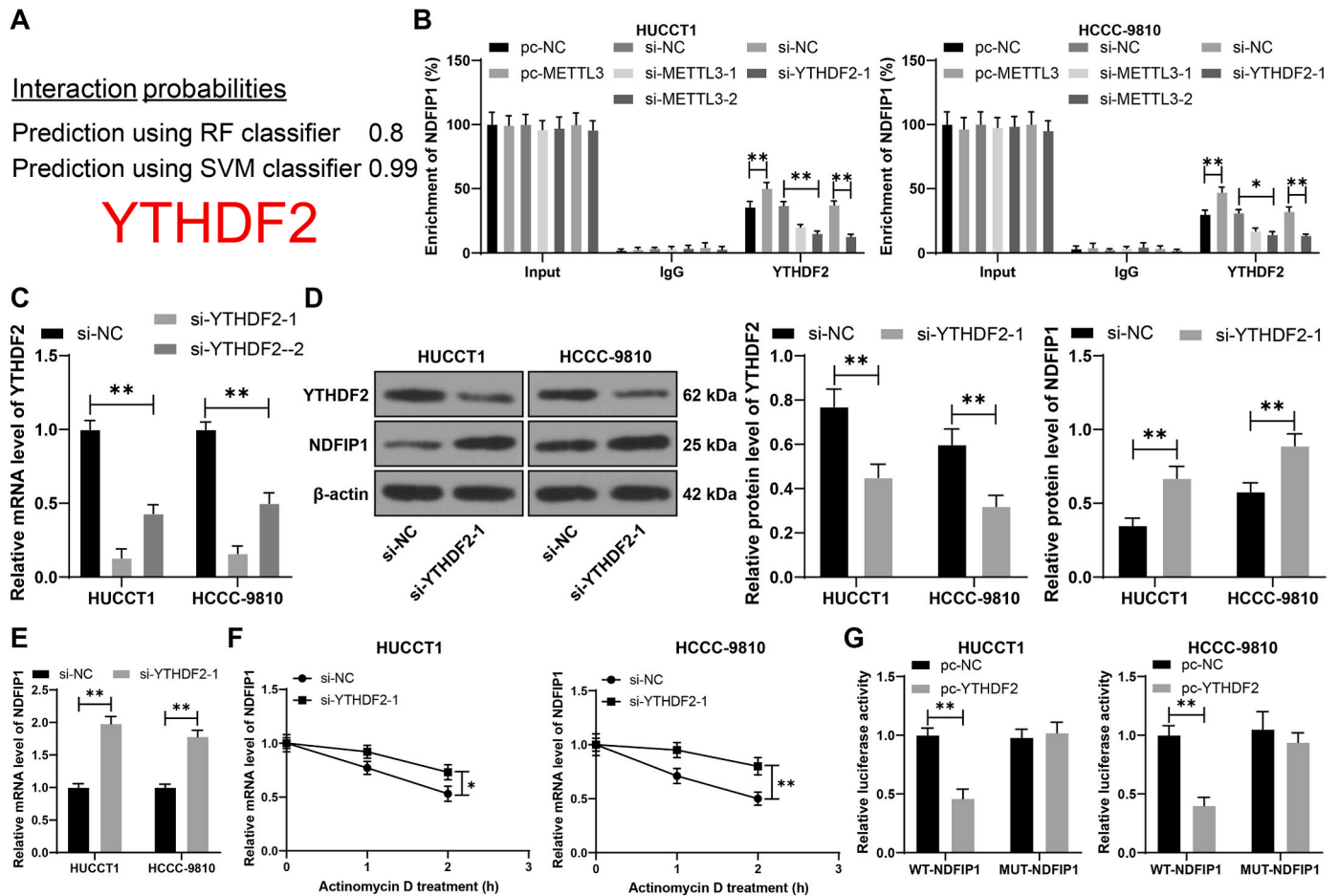


Fig. 4. YTHDF2 recognizes m6A and degrades mRNA to suppress NDFIP1 expression. **A:** The binding probability between YTHDF2 and NDFIP1 mRNA was predicted by the RPISeq database. **B:** RIP analysis of YTHDF2 enrichment on NDFIP1 mRNA in cells. **C:** The expression of YTHDF2 in different cells after siRNA transfection was detected by RT-qPCR. **D:** The expression of YTHDF2 and NDFIP1 in different cells was detected by Western blot analysis. **E:** The expression of NDFIP1 in cells was detected by RT-qPCR. **F:** After actinomycin D treatment, the stability of NDFIP1 mRNA in cells was assessed by RT-qPCR. **G:** The binding of YTHDF2 to methylated NDFIP1 mRNA was detected by dual-luciferase assay. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. Two-way ANOVA was used for data comparison among multiple groups in panels B–G. Tukey's multiple comparisons test was used for post hoc analysis. * $p < 0.05$, ** $p < 0.01$.

migrating cells ($p < 0.05$, Fig. 5C–E). These findings indicate that downregulation of NDFIP1 alleviates the inhibitory effects of METTL3 silencing on ICC cell proliferation, invasion, and migration.

3.6. Overexpression of NDFIP1 alleviates the promotive effects of METTL3 overexpression on ICC cell proliferation, invasion, and migration

Next, we upregulated NDFIP1 expression in HCCC-9810 cells by transfecting pc-NDFIP1 and performed combined experiments with pc-METTL3 ($p < 0.05$, Fig. 6A–B). The results demonstrated that after upregulating NDFIP1, cell proliferation, invasion, and migration were significantly reduced ($p < 0.05$, Fig. 6C–E). This result indicates that overexpression of NDFIP1 alleviates the promotive effects of METTL3 overexpression on ICC cell proliferation, invasion, and migration.

3.7. METTL3 promotes ICC cell proliferation in vivo

Finally, we established a xenograft tumor model in nude mice to validate the role and mechanism of METTL3. Upregulation of METTL3 increased tumor volume and weight and elevated Ki67-positive rate in tumor tissues, while downregulation of METTL3 suppressed tumor growth ($p < 0.05$, Fig. 7A–C). Nevertheless, upregulation or downregulation of NDFIP1 ($p < 0.05$, Fig. 7F–G) partially reverses the effect of METTL3 overexpression or METTL3 downregulation on tumor growth,

respectively ($p < 0.05$, Fig. 7A–C). Compared to the sh-NC group, the sh-METTL3 group showed decreased METTL3 expression, reduced m6A content, decreased m6A modification on NDFIP1, and upregulated NDFIP1 expression, whereas the oe-METTL3 group exhibited the opposite trend compared to the oe-NC group ($p < 0.05$, Fig. 7D–G). These findings indicate that METTL3 promotes ICC cell proliferation by decreasing NDFIP1 expression via m6A modification.

4. Discussion

Aberrant gene expression mediated by m6A methylation is a critical contributor to the malignant phenotype of ICC [25]. In this study, we found that the methyltransferase METTL3 promoted the degradation of NDFIP1 mRNA by increasing m6A modification on NDFIP1 mRNA and facilitating the binding of YTHDF2 to NDFIP1 mRNA, thereby decreasing NDFIP1 expression and finally promoting cell proliferation, invasion, and migration in ICC (Fig. 8).

The oncogenic role of METTL3 has been extensively reported [26, 27]. Notably, overexpression of METTL3 has the potential to induce proliferation, migration, and invasion of ICC cells [28]. Consistently, our study highlighted that METTL3 was highly expressed in ICC cell lines, and overexpression of METTL3 promoted proliferation, invasion, and migration of ICC cells, and *in vivo*, METTL3 promotes tumorigenesis. In contrast, downregulation of METTL3 inhibited the malignant phenotype

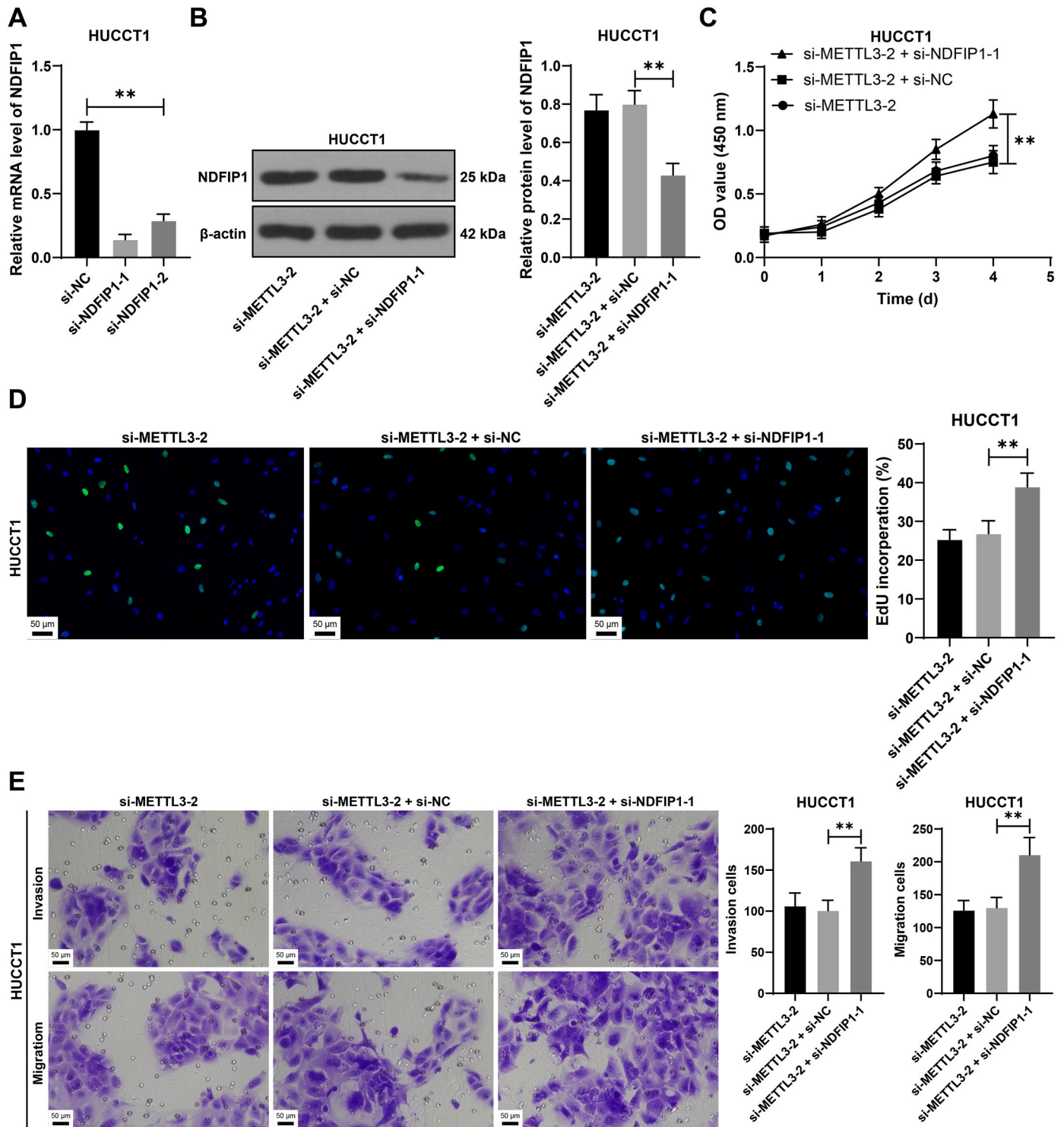


Fig. 5. Downregulation of NDFIP1 alleviates the inhibitory effects of METTL3 silencing on ICC cell proliferation, invasion, and migration. HUCCT1 cells were transfected with si-NDFIP1, with si-NC as the negative control. A–B: The expression of NDFIP1 in cells was detected by RT-qPCR and Western blot assays. C: Cell proliferation was measured using the CCK-8 assay. D: The number of EdU-positive cells was detected by EdU staining. E: Cell invasion and migration were assessed using the Transwell assay. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. One-way ANOVA was used for data comparison among multiple groups in panels A–B and D–E, while two-way ANOVA was used for data comparison in panel C. Tukey's multiple comparisons test was used for post hoc analysis. $**p < 0.01$.

of ICC cells and reduced tumor growth. METTL3 regulates downstream genes in various types of cancer in an m6A-dependent manner, including bladder cancer [29], small cell lung cancer [30], and papillary thyroid cancer [31]. Our further analysis revealed that METTL3 significantly increased the m6A content in ICC cells. Consistently, METTL3

upregulates m6A modification on NFAT5 and increases NFAT5 expression, resulting in increased proliferation and aggravated metastasis in ICC [32]. METTL3 facilitates HLF expression through the m6A methylation modification, leading to augmented proliferative and metastatic capacities of ICC cells [25]. In conclusion, METTL3 promotes ICC

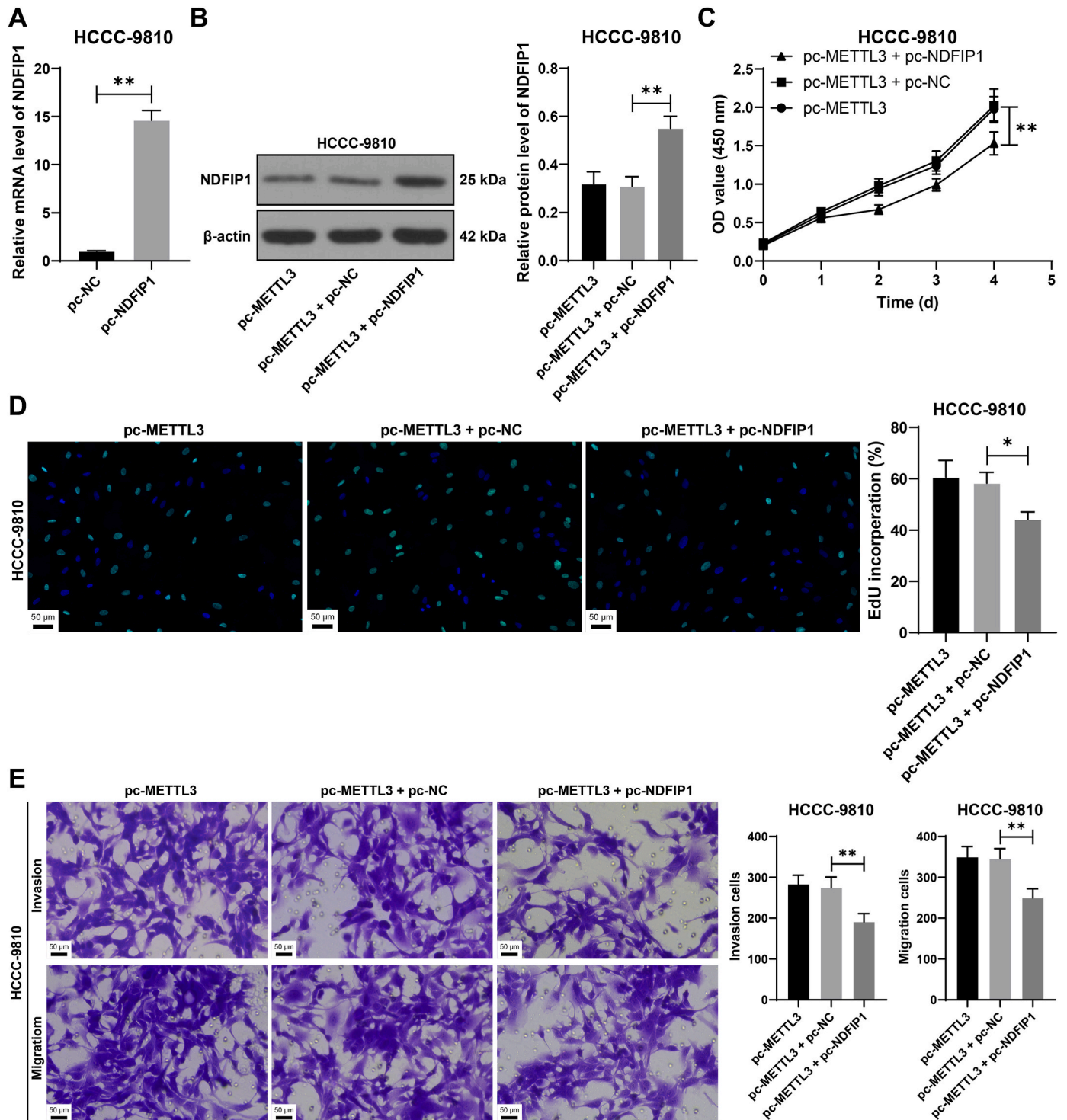


Fig. 6. Overexpression of NDFIP1 alleviates the promotive effects of METTL3 overexpression on ICC cell proliferation, invasion, and migration. HCCC-9810 cells were transfected with pc-NDFIP1, with pc-NC as the negative control. A–B: The expression of NDFIP1 in cells was detected by RT-qPCR and Western blot assays. C: Cell proliferation was measured using the CCK-8 assay. D: The number of EdU-positive cells was detected by EdU staining. E: Cell invasion and migration were assessed using the Transwell assay. The experiments were performed in triplicate, and the data are presented as the mean $\pm$ standard deviation. The *t*-test was used for data comparison between two groups in panel A. One-way ANOVA was used for data comparison among multiple groups in panels B, D, and E, while two-way ANOVA was used for data comparison in panel C. Tukey's multiple comparisons test was used for post hoc analysis. **p* < 0.05, ***p* < 0.01.

progression by increasing the content of m6A in cells.

METTL3-mediated m6A modification may be recognized by YTHDF2, leading to tumor suppressor mRNA degradation and down-regulation [33]. Elevated expression of YTHDF2 and METTL3 is observed in prostate cancer tissues, which increases m6A modification

levels and promotes tumor growth and metastasis through YTHDF2-mediated mRNA degradation [34]. Interestingly, another study indicates that upregulation of METTL3 increases m6A modification and accelerates the decay of IFIT2 mRNA in a YTHDF2-dependent manner, thus promoting the development of ICC [12]. In our study,

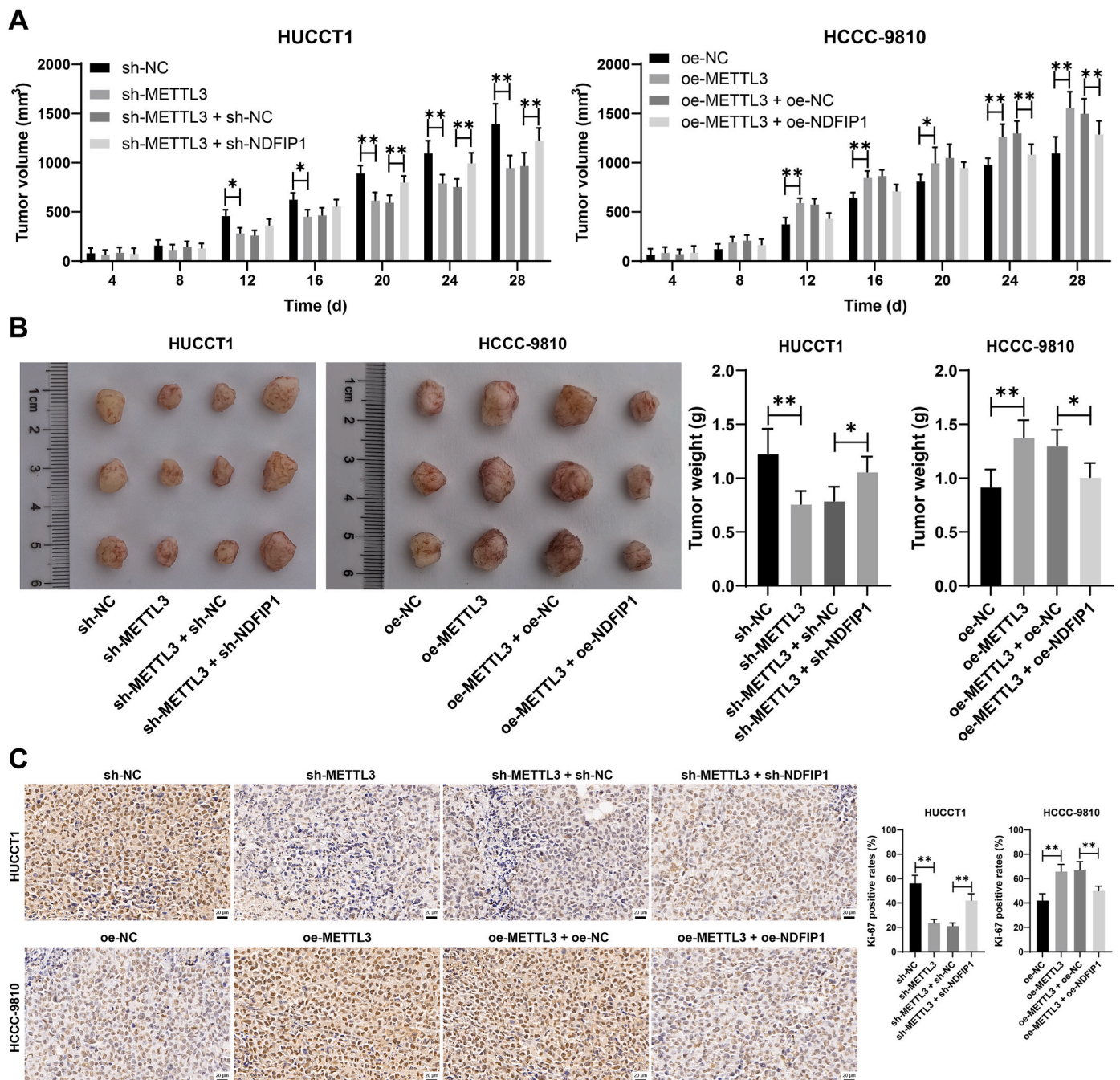


Fig. 7. METTL3 promotes ICC cell proliferation *in vivo*. A: Tumor size was measured every 4 days after subcutaneous tumor formation in nude mice. On day 28, the mice were euthanized, and tumor tissues were photographed. B: Tumor weight and representative images of tumor tissues. C: Immunohistochemical staining was performed to determine the Ki67-positive rate in tumor tissues. D: Quantitative analysis of m6A levels in tissues. E: The enrichment of m6A modification on NDFIP1 mRNA in the tissue was analyzed by RIP assay. F–G: The expression of METTL3 and NDFIP1 in tumor tissues was detected by RT-qPCR and Western blot assays. N = 6, and the experiments were performed in triplicate. The data are presented as the mean $\pm$ standard deviation. One-way ANOVA was used for data comparison among multiple groups in panels B–D and F–G, while two-way ANOVA was used for data comparison in panels A and E. Tukey's multiple comparisons test was used for post hoc analysis. * $p < 0.05$, ** $p < 0.01$.

we initially observed an increase in m6A modification level on NDFIP1 mRNA following the overexpression of METTL3, along with decreased NDFIP1 mRNA stability and expression. Additionally, METTL3 recruited YTHDF2 to recognize m6A and degrade NDFIP1 mRNA, consequently reducing NDFIP1 expression. In summary, METTL3-mediated m6A modification downregulates NDFIP1 expression in a YTHDF2-dependent manner.

Loss of NDFIP1 leads to dysregulation of PTEN (a tumor suppressor

protein) and results in abnormal cell proliferation [35]. An array of studies has documented the pivotal role of NDFIP1 as a key tumor suppressor [19,36]. For example, NDFIP1 is downregulated in breast cancer and has the potential to serve as a prognostic biomarker [37]. NDFIP1 expression is significantly decreased in uveal melanoma cells, while ectopic expression of NDFIP1 can counteract cancer cell proliferation and invasion [38]. Significantly, ubiquitination-mediated degradation of NDFIP1 protein levels can promote ICC proliferation,

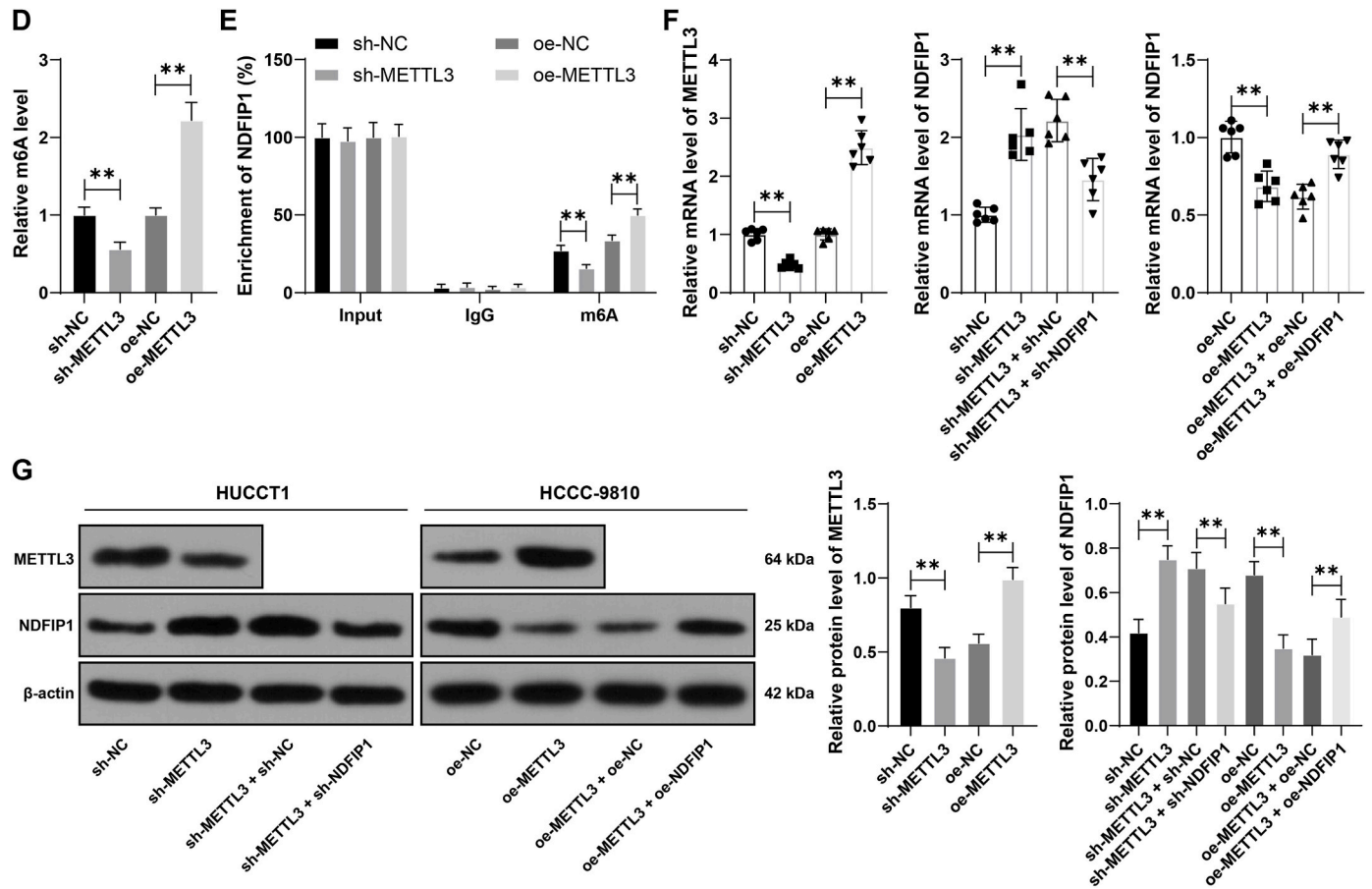


Fig. 7. (continued).

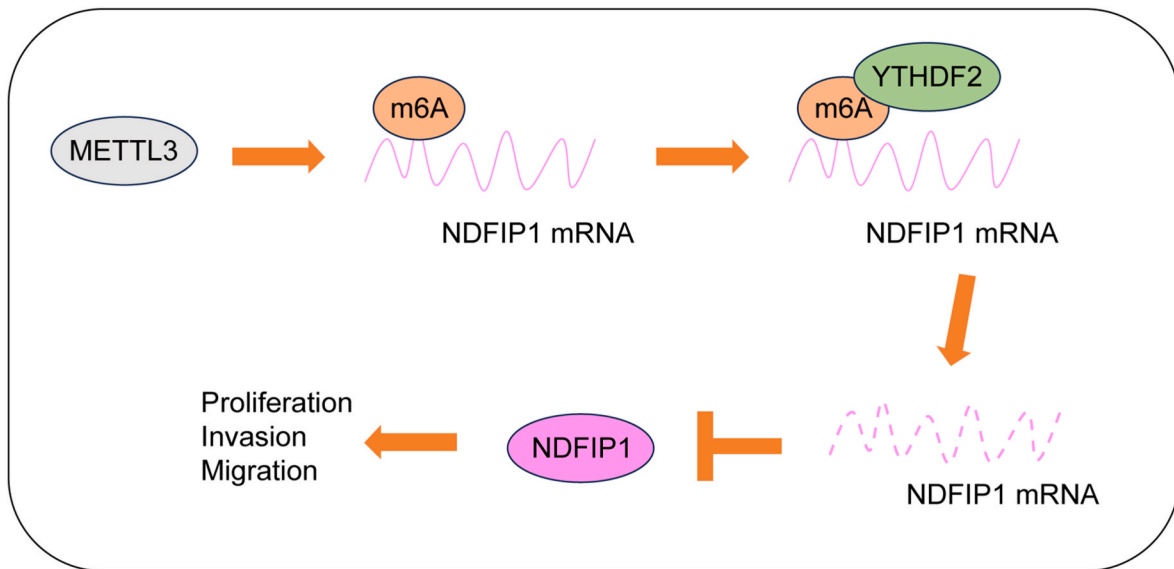


Fig. 8. Mechanism of METTL3 in promoting ICC cell proliferation, invasion, and migration. METTL3 promotes ICC cell proliferation, invasion, and migration by increasing m6A modification on NDFIP1 mRNA, which enhances the degradation of NDFIP1 mRNA in a YTHDF2-dependent manner.

migration, and invasion [20]. Our research findings further confirmed the suppressed expression of NDFIP1 in ICC. Overexpression of NDFIP1 significantly attenuated the proliferation, invasion, and migration of cancer cells. In contrast, NDFIP1 downregulation promoted the malignant phenotype of ICC cells. Taken together, as a tumor suppressor, NDFIP1 expression is inhibited in ICC, and NDFIP1 upregulation can

effectively inhibit the malignant phenotype of ICC cells.

There are still several limitations in this study. First, this study only investigated the downstream mechanisms of METTL3, and the upstream mechanisms of METTL3 have not been examined. Second, the effects of the single mechanism have only been explored in two cell lines, and whether different results would be observed in other cell types remains

to be investigated. Third, only the role of the m6A reader protein YTHDF2 has been examined, and the involvement of other reader proteins is still unclear. Meanwhile, m6A-seq and RNA-seq analyses were not conducted to elucidate the downstream targets and potential mechanisms of METTL3. Furthermore, our focus was limited to the observation of tumor size, and we did not assess METTL3 expression during the course of the experiments. Consequently, this study provides only a preliminary exploration of the role of METTL3 in ICC. In future studies, we will explore the upstream mechanisms of METTL3, investigate the mechanism in other cell lines, and explore the potential functions of other m6A reader proteins, providing new theoretical knowledge for the treatment of ICC.

In conclusion, METTL3 promotes ICC cell proliferation, invasion, and migration by increasing m6A modification on NDFIP1 mRNA and reducing NDFIP1 expression in a YTHDF2-dependent manner.

CRediT authorship contribution statement

Xinmiao Jiang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Hui Tan:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no conflicts of interest to disclose.

Data availability

Data will be made available on request.

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